

Wenbin Zhai

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PROFESSIONAL SUMMARY

Ph.D. student in the Department of Computing at The Hong Kong Polytechnic University.

My current research focuses on LLM agent security, especially tool-augmented LLM agents, agentic workflows, and external agent components such as tools, memory, and retrieval systems.

My broader research interests include cybersecurity, trustworthy AI, UAV/IoT networks, anomaly detection, and secure distributed systems.

EDUCATION

The Hong Kong Polytechnic University (PolyU), Hong Kong

Sep. 2025 – Present

Ph.D. in Computer Science and Technology, Department of Computing

Nanjing University of Aeronautics and Astronautics (NCAA), China

Sep. 2020 – Apr. 2023

M.Eng. in Computer Science and Technology, College of Computer Science and Technology

- GPA: 85.4/100, Ranking A (Top 15%)
- Thesis Topic: Research on Routing Protocol for Multi-hop Unmanned Aerial Vehicle Ad-hoc Networks (Outstanding Postgraduate Thesis of Jiangsu Province and Outstanding Postgraduate Thesis of Jiangsu Computer Society)

Nanjing University of Chinese Medicine (NJUCM), China

Sep. 2016 – Jun. 2020

B.Eng. in Computer Science and Technology, School of Artificial Intelligence and Information Technology

- GPA: 86/100, Ranking 8/61 (Professional GPA: 90.8/100, Ranking 2/61)
- Thesis Topic: Design and Implementation of a Method for the Safe Storage of Chinese Medicine Data based on Homomorphic Encryption (Outstanding Undergraduate Thesis of Jiangsu Province)

PUBLICATIONS

Journal Papers:

- **Wenbin Zhai**, Liang Liu, Youwei Ding, Shanshan Sun, and Ying Gu, "ETD: An Efficient Time Delay Attack Detection Framework for UAV Networks" in *IEEE Transactions on Information Forensics and Security (TIFS)*. [CORE A, CCF A, SCI Q1]
- **Wenbin Zhai**, Shanshan Sun, Liang Liu, Youwei Ding, and Wanying Lu, "HOTD: A Holistic Cross-Layer Time Delay Attack Detection Framework for UAV Networks" in *Journal of Parallel and Distributed Computing (JPDC)*. [CORE A, CCF B, SCI Q1]
- **Wenbin Zhai**, Liang Liu, Feng Wang, Youwei Ding, and Wanying Lu, "Federated Semi-Supervised and Semi-Asynchronous Learning for Anomaly Detection in IoT Networks" in *IEEE Transactions on Network and Service Management (TNSM)*. [CCF C, SCI Q2]
- Liang Liu*, **Wenbin Zhai***, Xin Li, Youwei Ding, Wanying Lu, and Ran Wang, "ESTA: An Efficient Spatial-Temporal Range Aggregation Query Processing Algorithm for UAV Networks" in *IEEE Transactions on Network Science and Engineering (TNSE)*. [CCF C, SCI Q1]
- Weichen Ding, **Wenbin Zhai**, Liang Liu, Ying Gu, and Hang Gao, "Detection of Packet Dropping Attack Based on Evidence Fusion in IoT Networks" in *Security and Communication Networks (SCN)*. [CCF C, SCI Q3]

- Kaibin Zhang, Liang Liu, **Wenbin Zhai**, Youwei Ding, and Jun Hu, "OSIS: Obstacle-Sensitive and Initial-Solution-First Path Planning" in *Peer-to-Peer Networking and Applications (PPNA)*. [CCF C, SCI Q2]
- Gongshun Min, Liang Liu, **Wenbin Zhai**, Zijie Wang, and Wanying Lu "An Efficient Data Collection Algorithm for Partitioned Wireless Sensor Networks" in *Future Generation Computer Systems (FGCS)*. [CORE A, CCF C, SCI Q1]
- Wenjie Zhao, Yu Wang, **Wenbin Zhai**, Liang Liu, and Yulei Liu, "Efficient Time-Delay Attack Detection Based on Node Pruning and Model Fusion in IoT Networks" in *Peer-to-Peer Networking and Applications (PPNA)*. [CCF C, SCI Q2]
- Yanlin Wang, Liang Liu, Mengqi Li, **Wenbin Zhai**, Weihua Ma, and Hang Gao, "Power Level Aware Charging Schedule in Wireless Rechargeable Sensor Network" in *Peer-to-Peer Networking and Applications (PPNA)*. [CCF C, SCI Q2]
- Yu Fan, Liang Liu, Xingxing Zhang, Huibin Shi, and **Wenbin Zhai**, "MAPP: An efficient multi-location task allocation framework with personalized location privacy-protecting in spatial crowdsourcing" in *Information Sciences*, 2023, 619: 654-678. [CORE A, CCF B, SCI Q1]
- Jiancheng Song, Liang Liu, Yulei Liu, Jie Xi, and **Wenbin Zhai**, "Path Planning for Multi-Vehicle-Assisted Multi-UAVs in Mobile Crowdsensing" in *Wireless Communications and Mobile Computing (WCMC)*, vol. 2022, 21 pages, 2022. [CCF C, SCI Q3]

Conference Papers:

- **Wenbin Zhai**, Liang Liu, Jianfei Peng, Youwei Ding, and Wanying Lu, "PAR: A Power-Aware Routing Algorithm for UAV Networks" in *17th International Conference on Wireless Algorithms, Systems, and Applications (WASA 2022)*, Dalian, China, November 24-26, 2022, Proceedings, Part III. Cham: Springer Nature Switzerland, 2022: 333-344. [CCF C]
- Yunfeng Cui, **Wenbin Zhai**, Liang Liu, Youwei Ding, and Wanying Lu, "A Framework for Moving Target Defense based on Federated Semi-Supervised Learning" in *3rd International Conference on Emerging Information Security and Applications (EISA 2022)*, Wuhan, China, October 29-30, 2022, Proceedings. Cham: Springer Nature Switzerland, 2023: 209-224.
- Wanying Lu, **Wenbin Zhai**, Feng Wang, and Yu Fan, "Link Aware Aggregation Query with Privacy-Preserving Capability in Wireless Sensor Networks" in *2023 International Conference on Electronics, Computers and Communication Technology (CECCT 2023)*, Guilin, China, November 17-19, 2023, Proceedings. Cham: Association for Computing Machinery, 2023: 245-250.
- Wanying Lu, Liang Liu, **Wenbin Zhai**, Haoyuan Chen, and Yulei Liu, "HBC: Combining Lossy and Lossless Hybrid Bilayer Compression Framework on Time-Series Data" in *21st IEEE International Symposium on Parallel and Distributed Processing with Applications (ISPA 2023)*, Wuhan, China, December 21-24, 2023. [CCF C]
- Kaibin Zhang, Liang Liu, **Wenbin Zhai**, Youwei Ding, and Jun Hu, "OSIS: Obstacle-Sensitive and Initial-Solution-First Path Planning" in *29th IEEE International Conference on Parallel and Distributed Systems (ICPADS 2023)*, Hainan, China, December 17-21, 2023. [CCF C]
- Kun Guo, Liang Liu, **Wenbin Zhai**, and Youwei Ding, "EKR: An Efficient K-anycast Routing in UAV Networks" in *9th International Conference on Computer and Communications (ICCC 2023)*, Chengdu, China, December 8-11, 2023.
- Zixiao Zhou, Liang Liu, **Wenbin Zhai**, Jiancheng Song, and Yulei Liu, "Power-Aware Path Planning for Vehicle-Assisted Heterogeneous UAVs in Mobile Crowd Sensing" in *2023 International Conference on Data, Information and Computing Science (CDICS 2023)*, Singapore, December 8-10, 2023.
- Keyue Yang, Liang Liu, Shijie Li, **Wenbin Zhai**, Wenjing Wang, Ziyi Zheng, and Jinan Wang, "DDC: Efficient Dynamic-Dictionary-Based Compression on Floating Time Series Data" in *23rd IEEE International Symposium on Parallel and Distributed Processing with Applications (ISPA 2025)*, Shenyang, China, October 10-12, 2025. [CCF C]

- Lingling Hu, Liang Liu, Yulei Liu, **Wenbin Zhai**, and Xinmeng Wang, "A Robust Fixed Path-Based Routing Scheme for Protecting Source Location Privacy in WSNs" in *17th International Conference on Mobility, Sensing and Networking (MSN 2021)*, Exeter, UK, December 13-15, 2021, IEEE, 2021: 48-55. [CCF C]

* Equal contribution / co-first authors.

SELECTED AWARDS AND HONORS

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|---|------------------------------|
| • PolyU Research Postgraduate Scholarship (2025) | <i>May. 2025</i> |
| • Outstanding Postgraduate Thesis of Jiangsu Province (2023) | <i>Dec. 2024</i> |
| • Outstanding Postgraduate Thesis of Jiangsu Computer Society (2023) | <i>Dec. 2024</i> |
| • Tuition Fee Scholarship of UNSW Sydney (2023) | <i>Dec. 2023</i> |
| • Provincial Merit Student of Jiangsu Province (2022-2023) | <i>Apr. 2023</i> |
| • Outstanding Graduate of NUAA (2023) | <i>Apr. 2023</i> |
| • Merit Student of NUAA (2021-2022) | <i>Dec. 2022</i> |
| • Advanced Individual in Research and Innovation of NUAA (2021-2022) | <i>Dec. 2022</i> |
| • Second-Class Scholarship for Postgraduate Students of NUAA (2020-2023) | <i>Sep. 2020 – Sep. 2022</i> |
| • Outstanding Undergraduate Thesis of Jiangsu Province (2020) | <i>Oct. 2021</i> |
| • Outstanding Graduate of NJUCM (2020) | <i>Jun. 2020</i> |
| • Merit Student of NJUCM (2018-2019) | <i>Dec. 2019</i> |
| • Ruihua Soaring Scholarship (2018-2019) | <i>Sep. 2019</i> |
| • National Encouragement Scholarship in China (2017-2019) | <i>Dec. 2018 – Dec. 2019</i> |
| • First-Class Scholarship for Undergraduate Students of NJUCM (2017-2019) | <i>Dec. 2018 – Dec. 2019</i> |
| • Principal's Special Award of NJUCM (2017-2018) | <i>Dec. 2018</i> |
| • Fei Xiaotong Virtue Scholarship (2017-2018) | <i>Dec. 2018</i> |

PROFESSIONAL ACTIVITIES

Conference Reviewer:

- The Fourteenth International Conference on Learning Representations (**ICLR 2026**) [CCF A]
- The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2026 (**CVPR 2026**) [CCF A]
- The Forty-Third International Conference on Machine Learning (**ICML 2026**) [CCF A]

Journal Reviewer:

- IEEE Transactions on Information Forensics and Security (**TIFS**) [CORE A, CCF A, SCI Q1]
- The Journal of Supercomputing (**TJSC**) [CCF C, SCI Q2]
- Frontiers of Information Technology & Electronic Engineering (**FITEE**) [CCF C, SCI Q2]